

Drivers of tie formation

Networks and Innovation Training Programme
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September 12, 2012

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People help
colleagues at work
for their own benefit.

People help
colleagues at work
for altruistic reasons.

“An apparent **altruism** pervades social life; people are anxious to reciprocate for the benefits they receive.

But beneath this seeming selflessness an underlying “egoism” can be discovered;

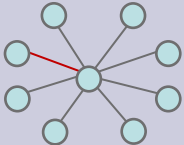
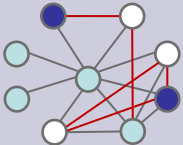
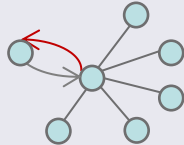
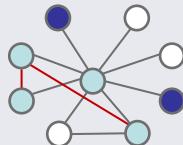
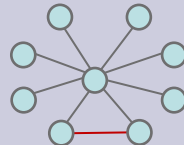
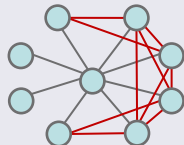
the tendency to help others is frequently motivated by the expectation that doing so will bring social returns”

Learning objectives

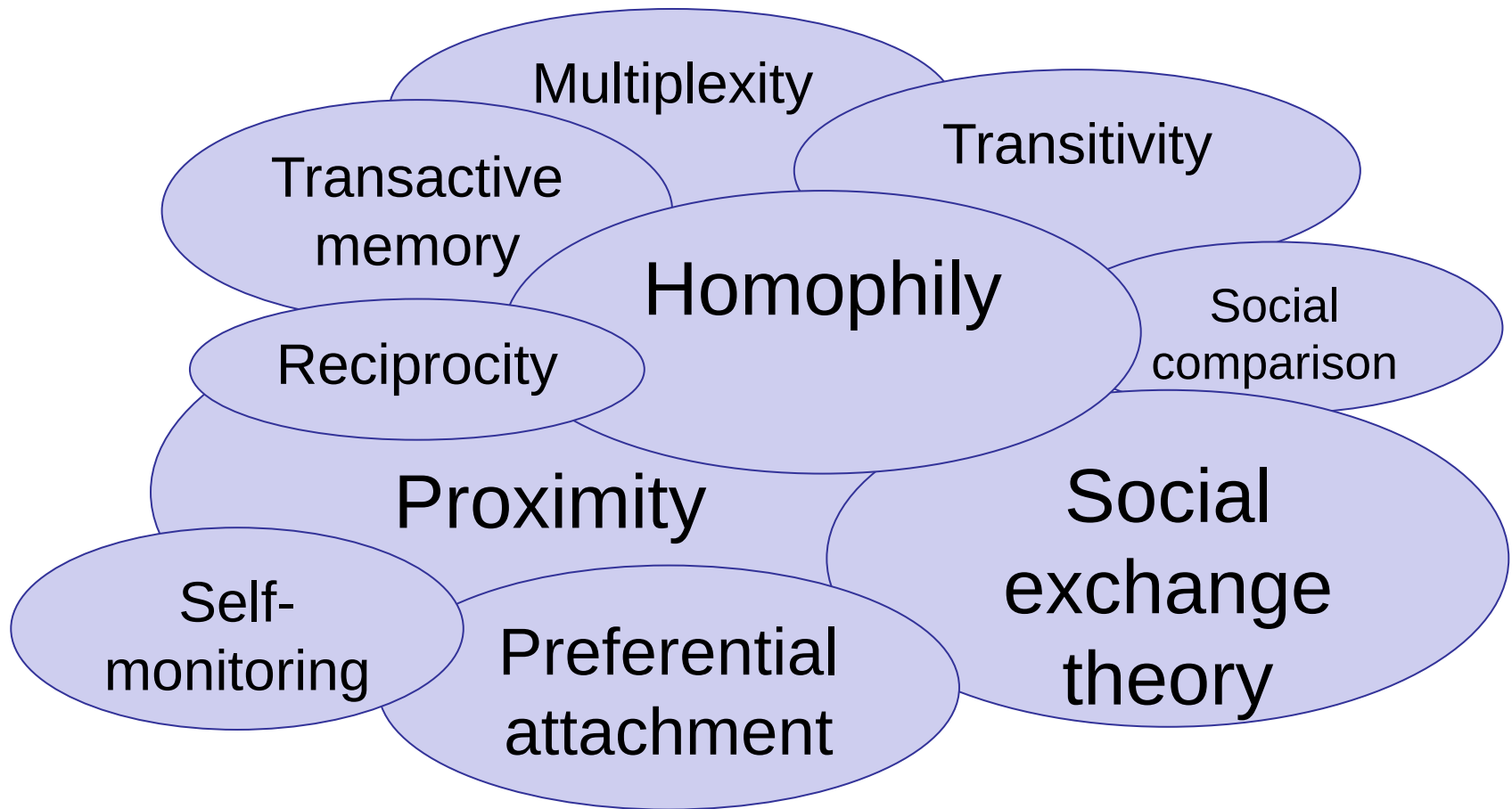
At the end of this lecture, we hope you will be able...

- ...distinguish and classify a range of **mechanisms** that may underlie the formation of network ties
- ...relate these mechanisms to the wider **sociological principles and theories** that underpin them
- ...understand the importance of **contextual factors** for the appropriate conduct of tie formation research

A typology of tie formation mechanisms

	Endogenous effects	Exogenous effects
Actor level		
Dyadic level		
Triadic level		
Network level		

Concepts



Endogenous mechanisms

- Networks are **path dependent**
- Network ties are like the weather:

“The best prediction of the network of tomorrow is the network of today”.

- The network of current ties set opportunities and constraints for the formation of new ties.
- Networks themselves form the **transmission mechanism** with information about the availability, suitability and reliability of potential partners.

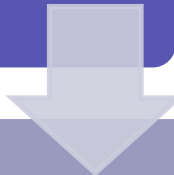
Social exchange theory

THEORETICAL PERSPECTIVES

Social attraction is the force that induces human beings to establish social associations on their own initiative.



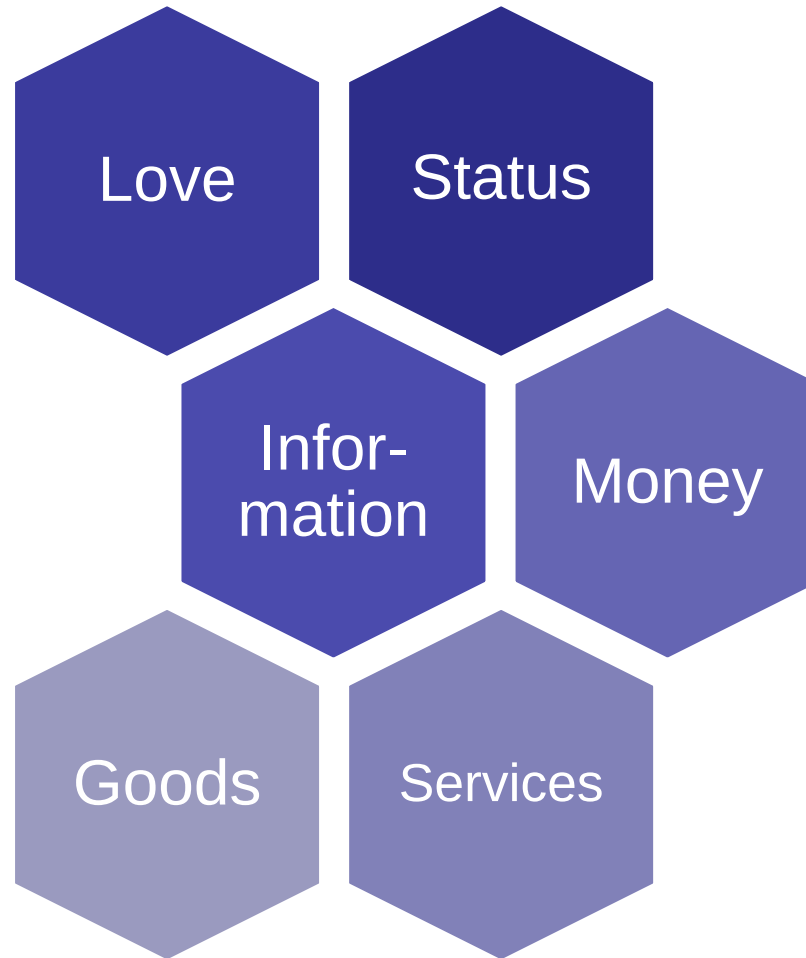
Forces of social attraction stimulate exchange transactions



Social exchange, in turn, tends to give rise to differentiation of status and power

Social exchange theory

THEORETICAL PERSPECTIVES



Source: Foa & Foa (1974), Cropanzano et al. (2006)

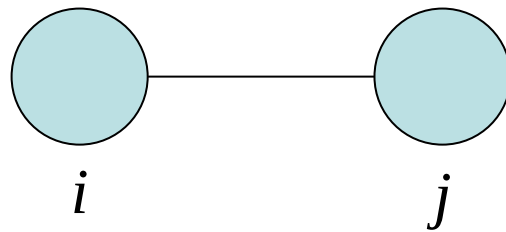
Social exchange theory

THEORETICAL PERSPECTIVES

The basic premise:

Individuals choose their social relationships evaluating the **cost** [what they give in the relation] and the **benefits** [what they receive from the relation] of the interaction against a set of **alternatives**.

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If both i and j take this approach to tie formation...
Does a tie between the two emerge? How?

Repetition and reciprocity

ENDOGENOUS DYADIC MECHANISMS

Social exchange involves a series of interactions that generate obligations.

Exchange interactions are usually seen as interdependent and contingent on the actions of another person.

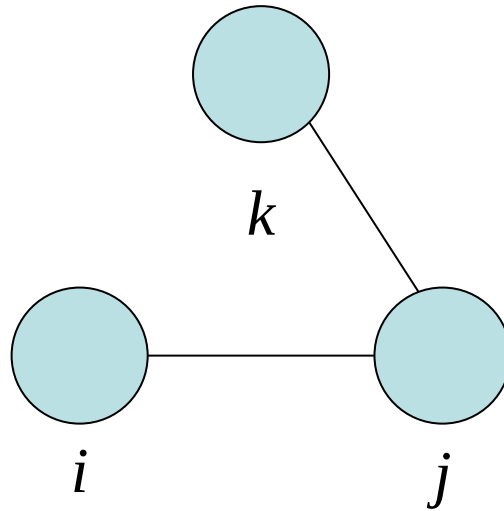
Repetition

Relationships evolve over time into trusting, loyal, and mutual commitments.

Reciprocity

Relationships are governed by rules of exchange, for example the expectation of reciprocity

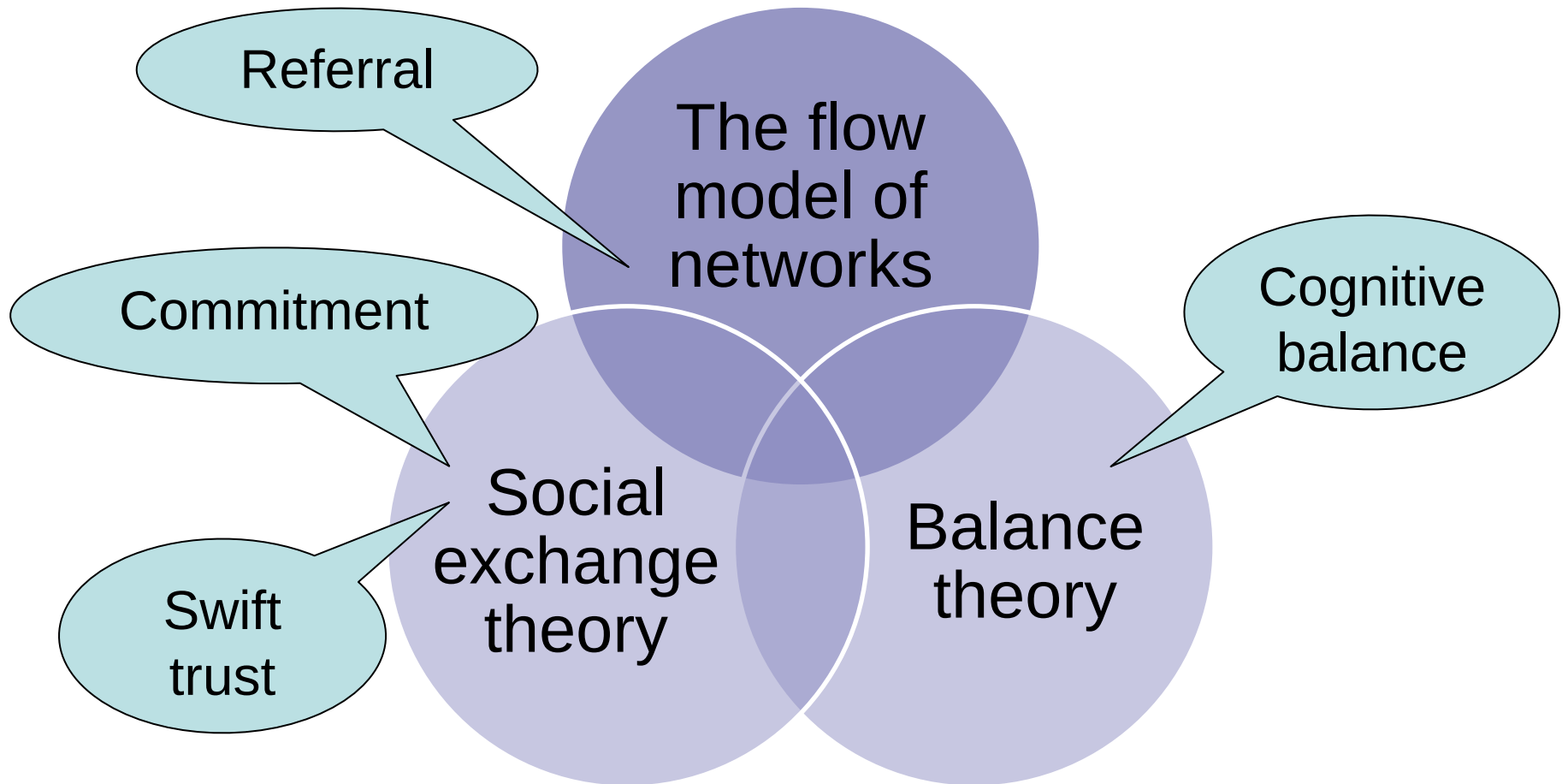
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If both i, j and k all take this approach to tie formation...
Why would a tie between j and k emerge?

Transitivity

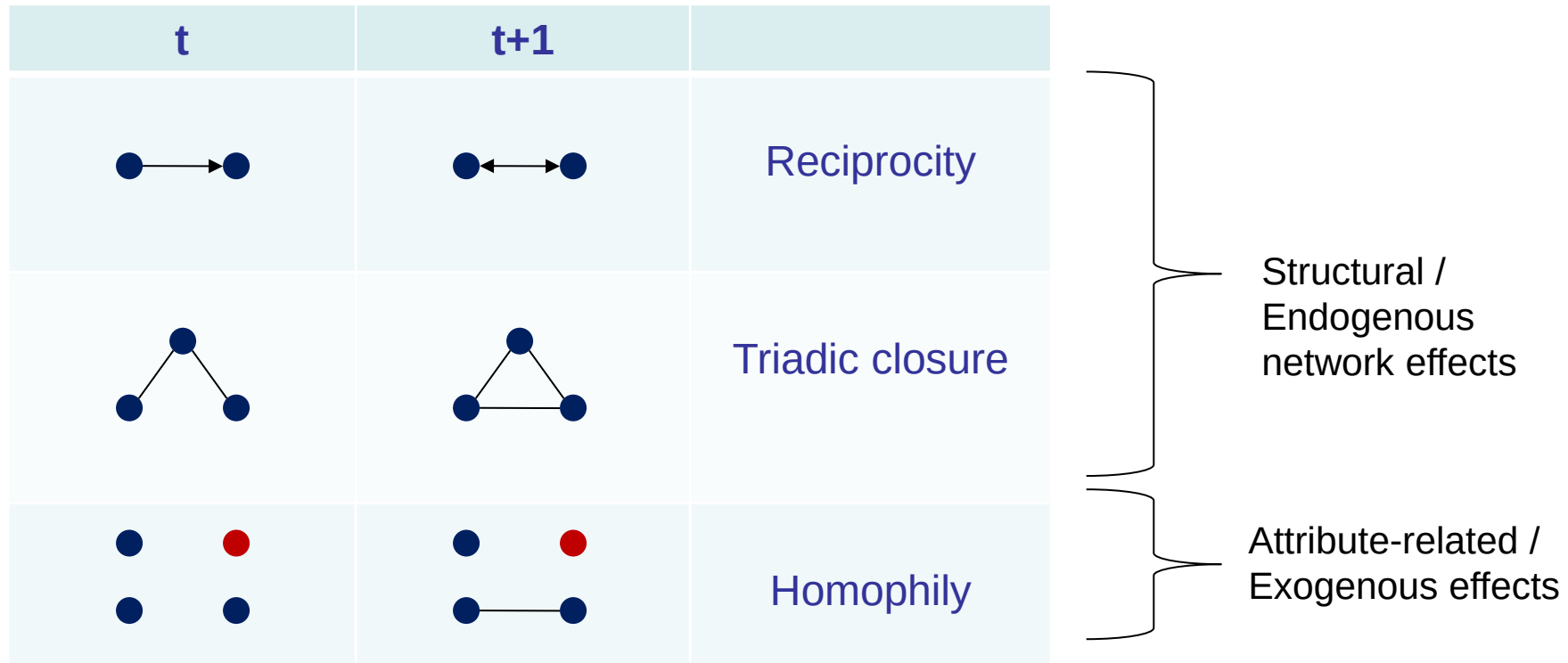
ENDOGENOUS TRADIC MECHANISMS



Social exchange theory

THEORETICAL PERSPECTIVES

Individuals try to create, maintain or restore balance in their personal relations.



Multiplexity

ENDOGENOUS NETWORK LEVEL MECHANISMS

The presence of a tie in one type of network may trigger the creation of a type in another.

The tendency to form relations in one **setting** can therefore only be fully understood in terms of which relations are present or absent in other settings.

- **Personal ties** may be a precursor to the emergence of economic exchange ties.
- **Economic exchange ties** may give rise to the development of personal ties.

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Instrumental ties
emerge from individuals'
deliberate strategic
decisions.

Instrumental ties
emerge from individuals'
unconscious, implicit
preferences for
interaction.

Homophily

EXOGENOUS DYADIC MECHANISMS



**“Birds of a feather
flock together”.**

Source: McPherson, Smith-Lovin & Cook (2001)

Homophily: the origins

EXOGENOUS DYADIC MECHANISMS

Assortative friendship processes:

- **Homophily**: tendency for friendships to form between people who are alike in some designated respect.
- **Heterophily**: opposite tendency.

Value homophily

correspondence in values of friends.

Status homophily

similarity of group affiliations



Social comparison theory

THEORETICAL PERSPECTIVES

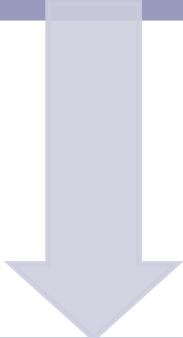
Humans have a drive to evaluate their opinions and abilities.



They do so by comparing with the opinions and abilities of others.



The tendency for comparison decreases as the difference between own and others abilities or opinions increases.



If a discrepancy exists

Change opinion

Influence others

Stop comparing

Most importantly: people make their lives easier and tend to associate with others with abilities and opinions near their own.

Homophily: overview

EXOGENOUS DYADIC MECHANISMS

Homophily
(McPherson et al. 2001)

Status homophily

(e.g. Lazega 2001; Rollag 2004;
Zenger & Lawrence 1989)

Ethnicity

Tenure

Gender

Value homophily

(e.g. Bock et al. 2001;
Osterloh & Frey 2000)

Attitudes
Beliefs

Homophily: how strong?

EXOGENOUS DYADIC MECHANISMS



Do people mix at mixers? Not really...



- Despite their goal to meet new people they overwhelmingly went back to pre-mixer friends.
- For new connections there was no evidence of homophily.
- Yet, they were more likely to join groups of people with at least one person of the same race.

Source: Ingram & Morris (2007)

Homophily: how strong?

EXOGENOUS DYADIC MECHANISMS

Choice homophily

Induced homophily



“The focused organization
of social ties”

Homophilous ties are formed
above what could be expected
at random given
the diversity of the setting.

Some social environments pose a
restricted opportunity structure
to form heterophilous ties.

Source: Feld (1981), McPherson & Smith-Lovin (1987)

Homophily: how strong?

EXOGENOUS DYADIC MECHANISMS



Longitudinal study of a
university community

Homophily is strong, but largely
the result of induced homophily:

- **triadic closure**
- **focal closure.**

Modest preferences for
similar others are **amplified**
through a cumulative process,
producing **striking patterns**
of observed homophily.

But...

**...homophily theory assumes
that people make unconscious
decisions who to associate with**

**What about the other “school”?
Creating ties with strategic intent.**

Homophily

EXOGENOUS DYADIC MECHANISMS

When building instrumental ties in organizations, individuals may deviate from choices informed by homophily for strategic reasons.

Lawrence (1997)

The homophily framework is somewhat at odds with more recent attempts to conceptualize networks more as strategic devices

Ibarra et al. (2005)

Complementarity

EXOGENOUS ACTOR-LEVEL MECHANISMS

Transactive memory theory

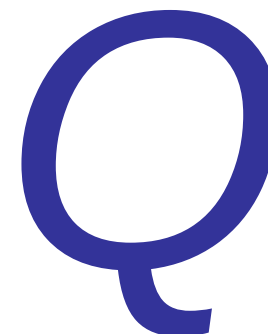
Instrumental ties (information seeking) in organizations is a function of:



Source: Borgatti & Cross (2003)

Personality

EXOGENOUS ACTOR-LEVEL MECHANISMS



I find it hard to imitate the behaviour of other people.

At parties and social gatherings, I do not attempt to do or say things that others will like.

I can only argue for ideas which I already believe in.

I can make impromptu speeches even on topics about which I have almost no information.

I guess

I would

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I would

I have considered being an entertainer.

I have never been good at games like charades or improvisational acting.

I have trouble changing my behaviour to suit different people and different situations.

At a party I let others keep the jokes and stories going.

I feel a bit awkward in public and do not show up quite as well as I should.

I can look anyone in the eye and tell a lie with a straight face (if for a right end).

I may deceive people by being friendly when I really dislike them.

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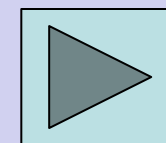
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“Why would this be related to tie formation?”



Source: Snyder & Gangestad (1986)

Personality

EXOGENOUS ACTOR-LEVEL MECHANISMS

Low self-monitors



True-to-themselves low self-monitors present themselves consistently in different social situations.

High self-monitors



Chameleon-like self-monitors adapt their self-presentation to the demands of different groups or situations.

→ Helps brokerage

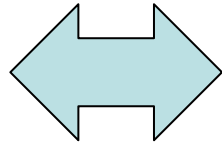
Networking strategy

EXOGENOUS ACTOR-LEVEL MECHANISMS

Catalysing strategies

NOT

Just-in-time building of strong ties.

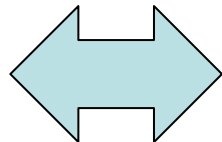


Casual-dating

Activating indirect or weak ties and transform them in strong ties.

NOT

Timing around resource needs.

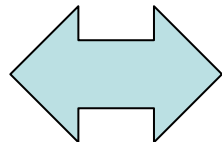


Timing around proof-points

Having tie attempts coincide with recent achievements or miles

NOT

Shopping for ties / Saying yes to everyone

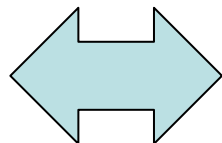


Scrutinizing interest

Weakening or dropping ties to trigger interest from third-parties

NOT

Naively hoping for ties

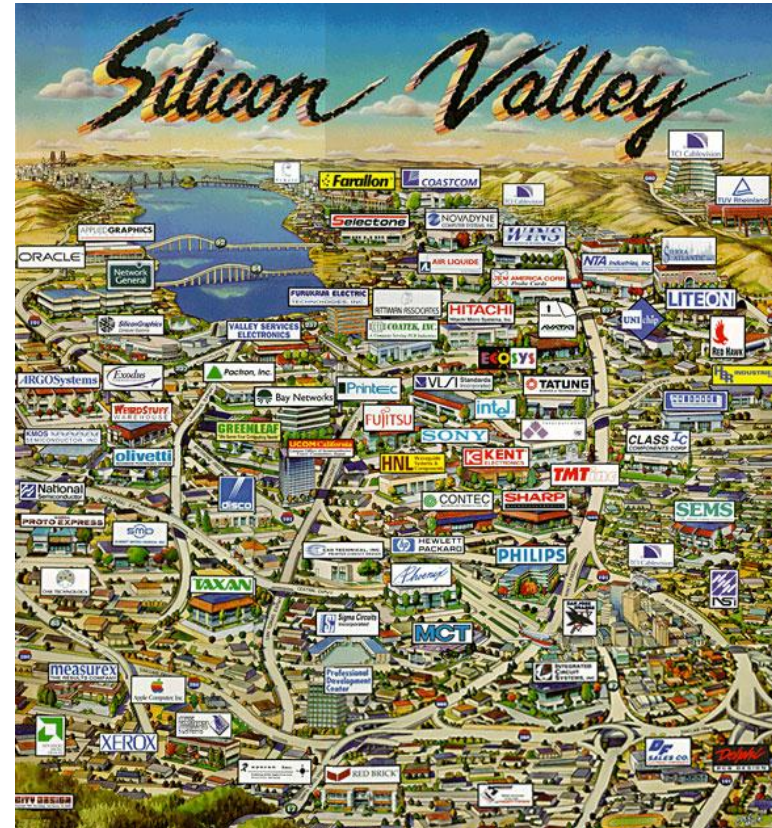
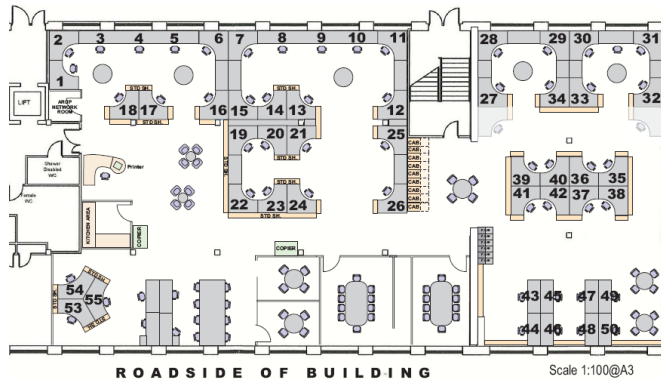


Crafting alternatives

May induce interested but hesitant partners to commit

Contextual factors

EXOGENOUS DYADIC MECHANISMS



Learning objectives

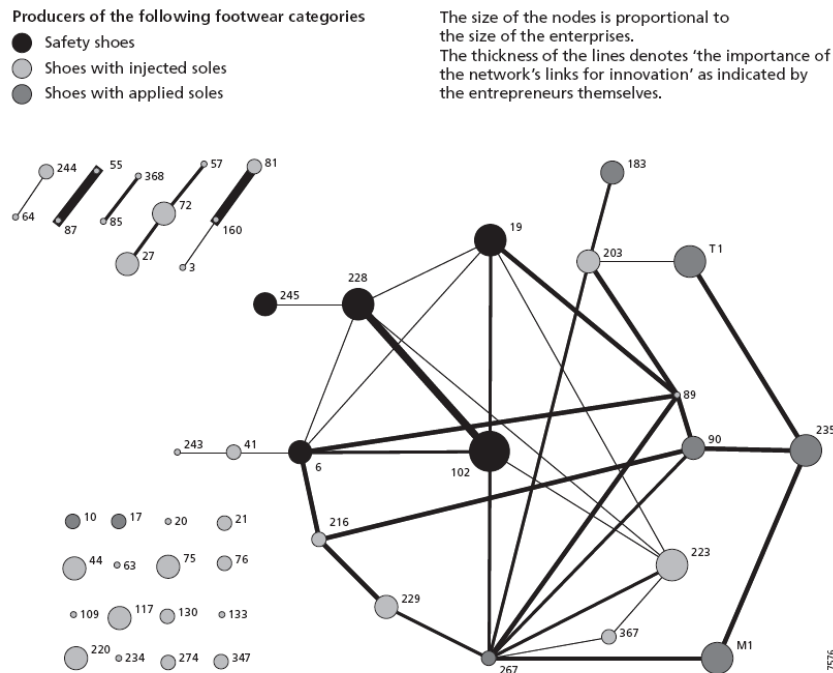
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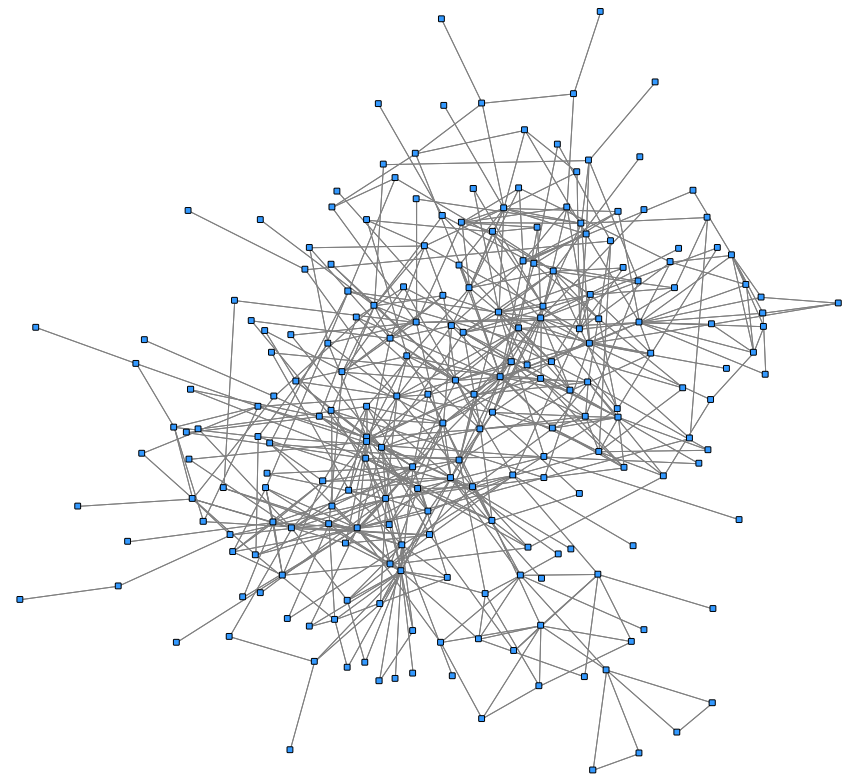


The importance of context and strategic intent in tie formation

Networks and clusters



Intra-organizational networks

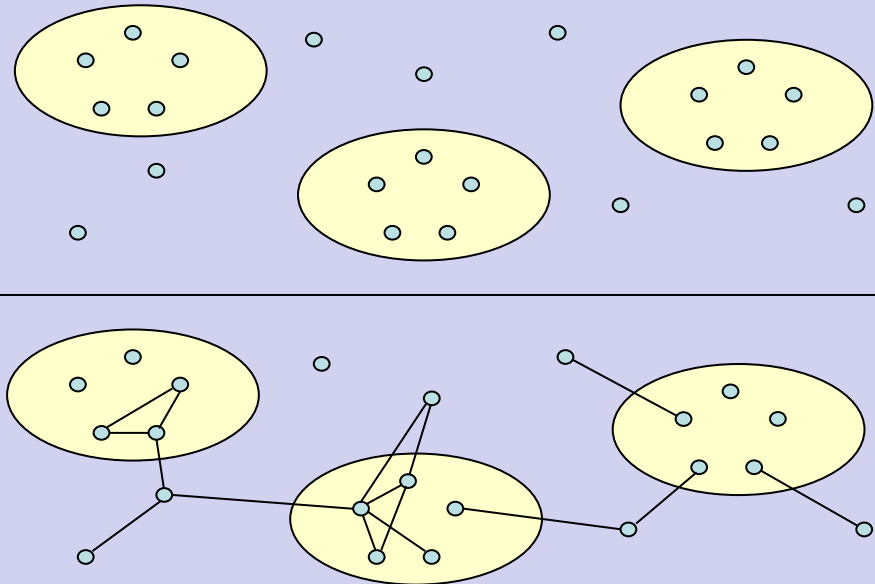


Source: Boschma & Ter Wal (2007), Criscuolo, Salter & Ter Wal (2013, WP)

Networks and clusters

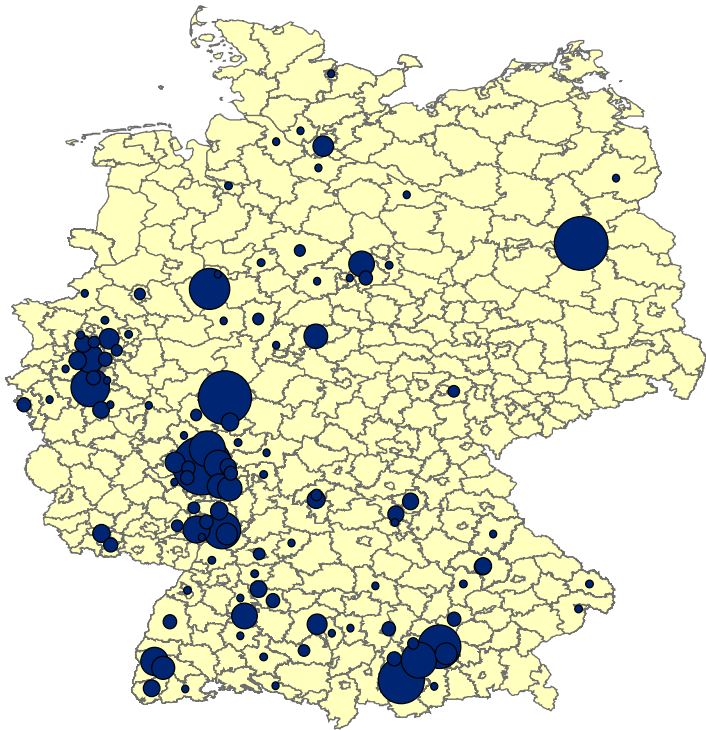
EXOGENOUS DYADIC MECHANISMS

Networks and clusters are two essential ways of getting access to new ideas.



Networks and clusters

EXOGENOUS DYADIC MECHANISMS



A **cluster** is a spatial concentration of organizations involved in related activities.

Clusters **may but need not** exhibit dense networks in which knowledge is transferred across organizations:

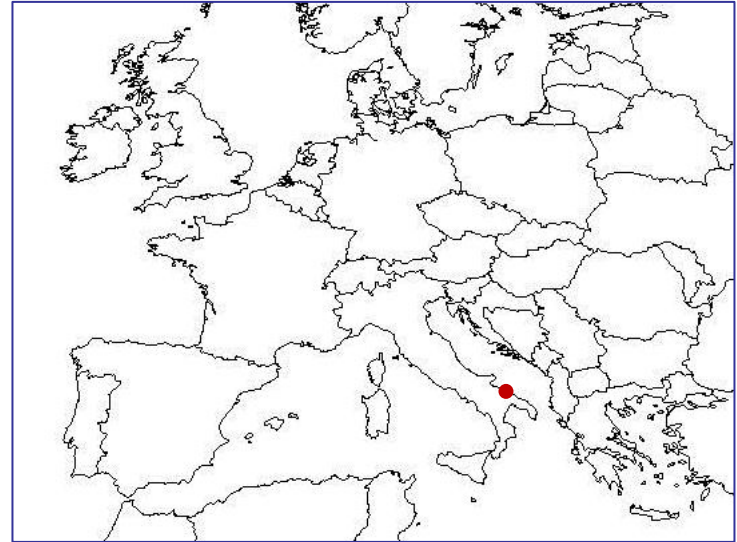
- Inter-firm collaboration
- Informal communities
- Labour mobility flows

• 1
• 10
• 100
Number of inventors per district (1991-1995) in German biotechnology

Networks and clusters

EXOGENOUS DYADIC MECHANISMS

Under what conditions do networks in clusters exist?



i.e. networks of collaboration that cross the boundaries of firms in a cluster

i.e. individual 'inventors' working for different companies that form a local community of 'collective learning' in which they share knowledge and collaborate.

Networks and clusters

EXOGENOUS DYADIC MECHANISMS



58 shoe producers

Around 70 specialized suppliers

Largest firm has around 300 employees

Most firms have around 20 employees

Firms largely differ in their R&D activity

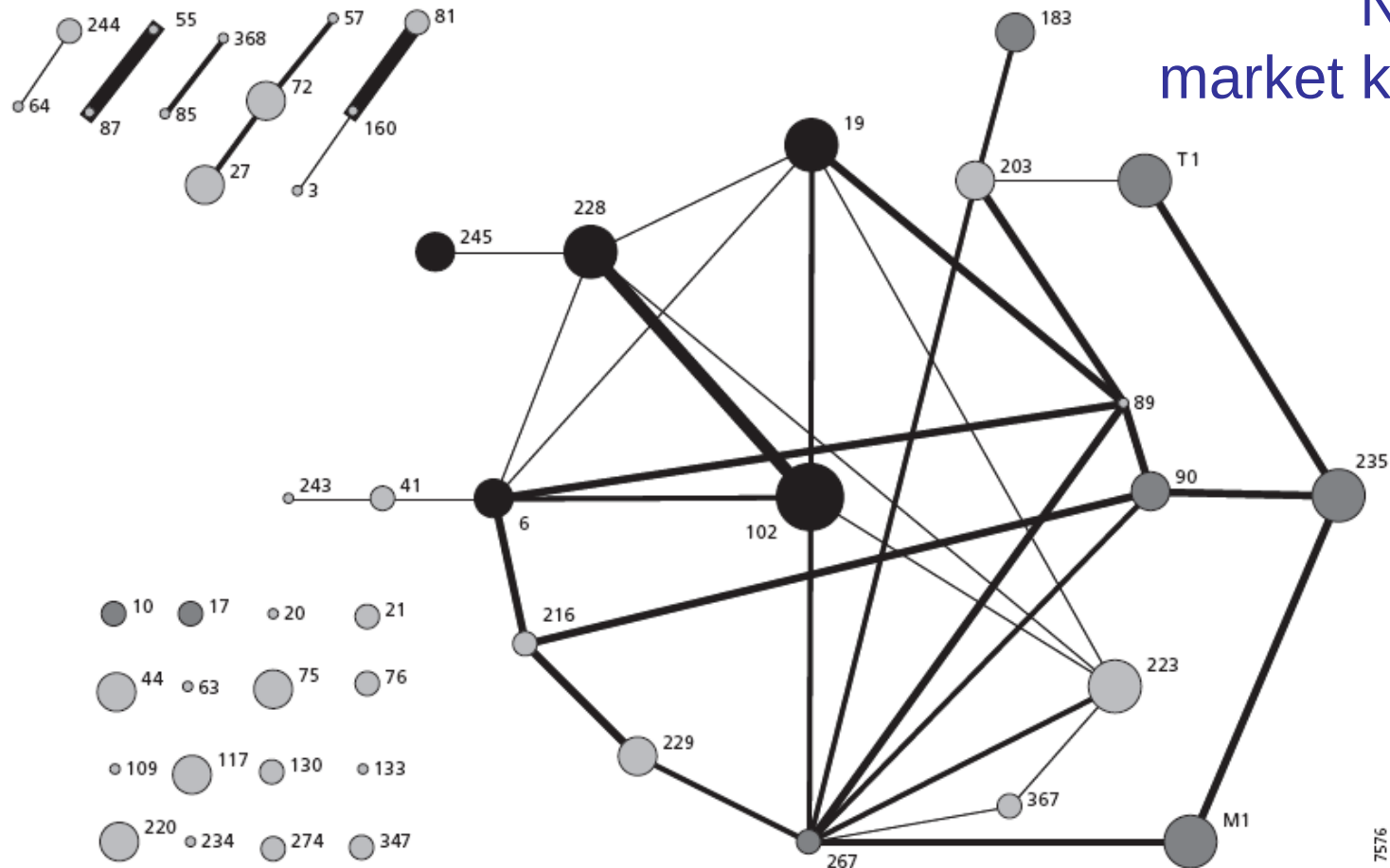


Source: Boschma & Ter Wal (2007)

Producers of the following footwear categories

- Safety shoes
- Shoes with injected soles
- Shoes with applied soles

The size of the nodes is proportional to the size of the enterprises.
The thickness of the lines denotes 'the importance of the network's links for innovation' as indicated by the entrepreneurs themselves.



Network of market knowledge

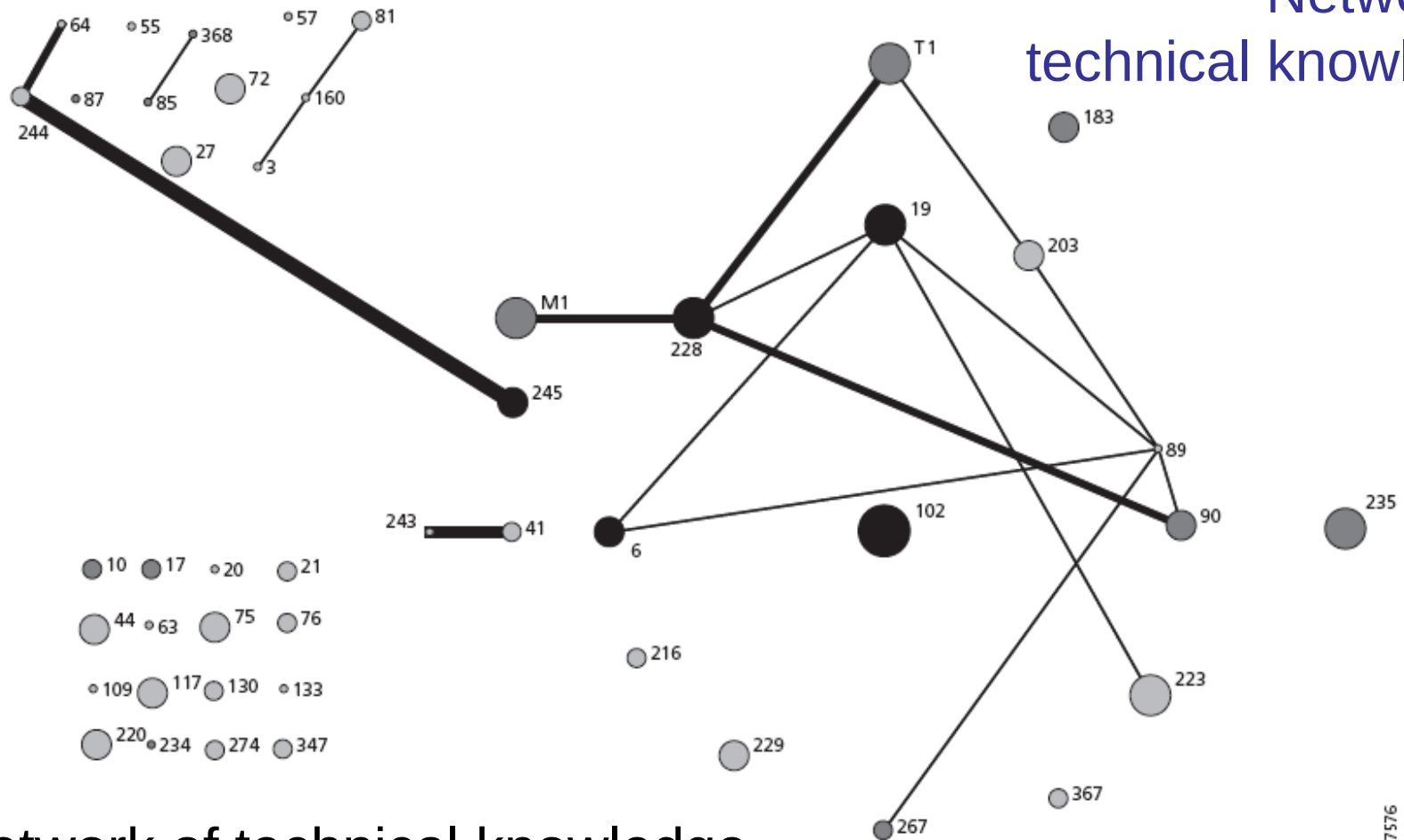
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Network of technical knowledge

Network of technical knowledge

Source: Boschma & Ter Wal (2007)

Networks and clusters

EXOGENOUS DYADIC MECHANISMS

Social explanations:

- Feelings of mistrust, due to intense competition in a mature market.
- Lack of social capital in the region.

Cognitive explanations:

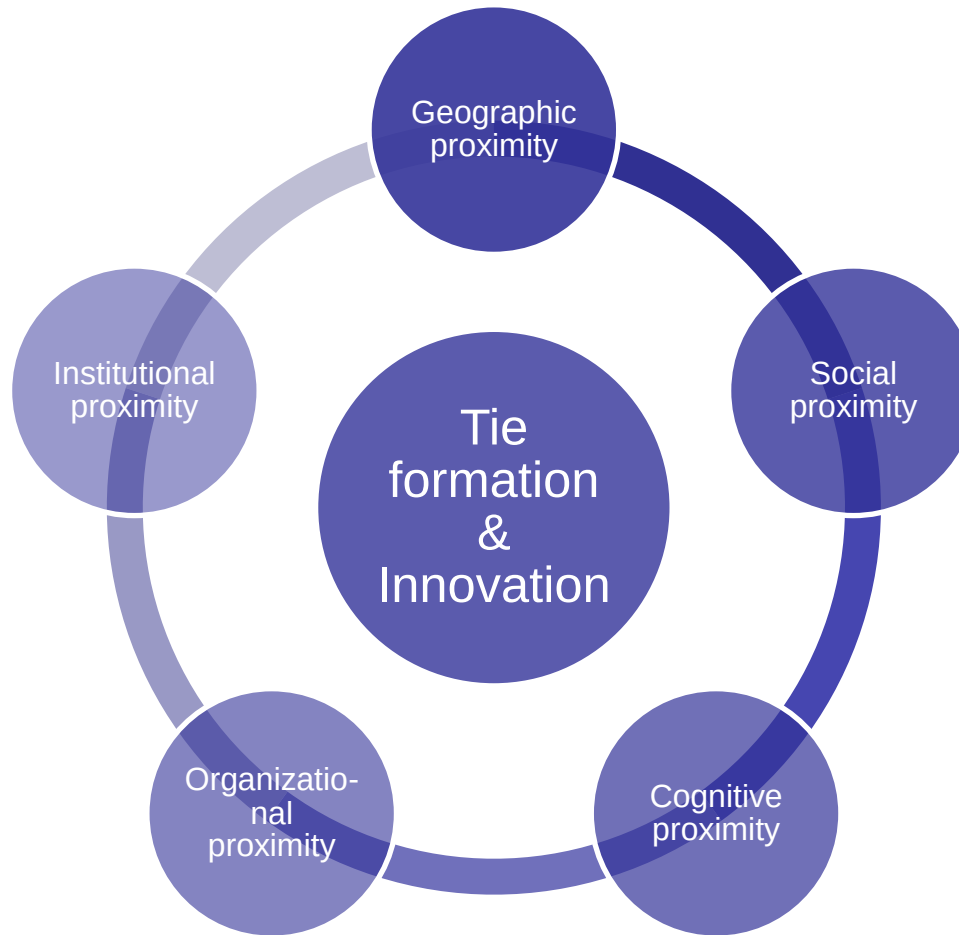
- Firms with stronger knowledge base are more centrally positioned in the network.
- Specialization of firms affect tie formation.

Geographical explanations:

- Non-local ties are missing in the picture.

Proximity

EXOGENOUS DYADIC MECHANISMS



Source: Boschma (2005)

Proximity

EXOGENOUS DYADIC MECHANISMS

GEO – The extent to which individuals are physically proximate.

Geography,
Spatial economics

SOC – The extent to which actors are embedded in a high-trust environment.

Sociology

COG – The extent to which actors share knowledge, skills and expertise

Cognition theory,
Learning

ORG – The extent to which actors share membership to organizational arrangements

Management &
Organization

INST – The extent to which two actors share norms, habits, routines

Institutional theory

Proximity

EXOGENOUS DYADIC MECHANISMS

The proximity typology is an **analytical toolbox** to **disentangle** the effects of various **mechanisms** on interaction.

Forms of proximity can **substitute** or **complement** one another.

Each form of proximity comes with a **long theoretical history**.

Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS



Source: Criscuolo, Salter & Ter Wal (2013, WP)

Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS



granite fountains



tunnel ventilation



deep basement excavation



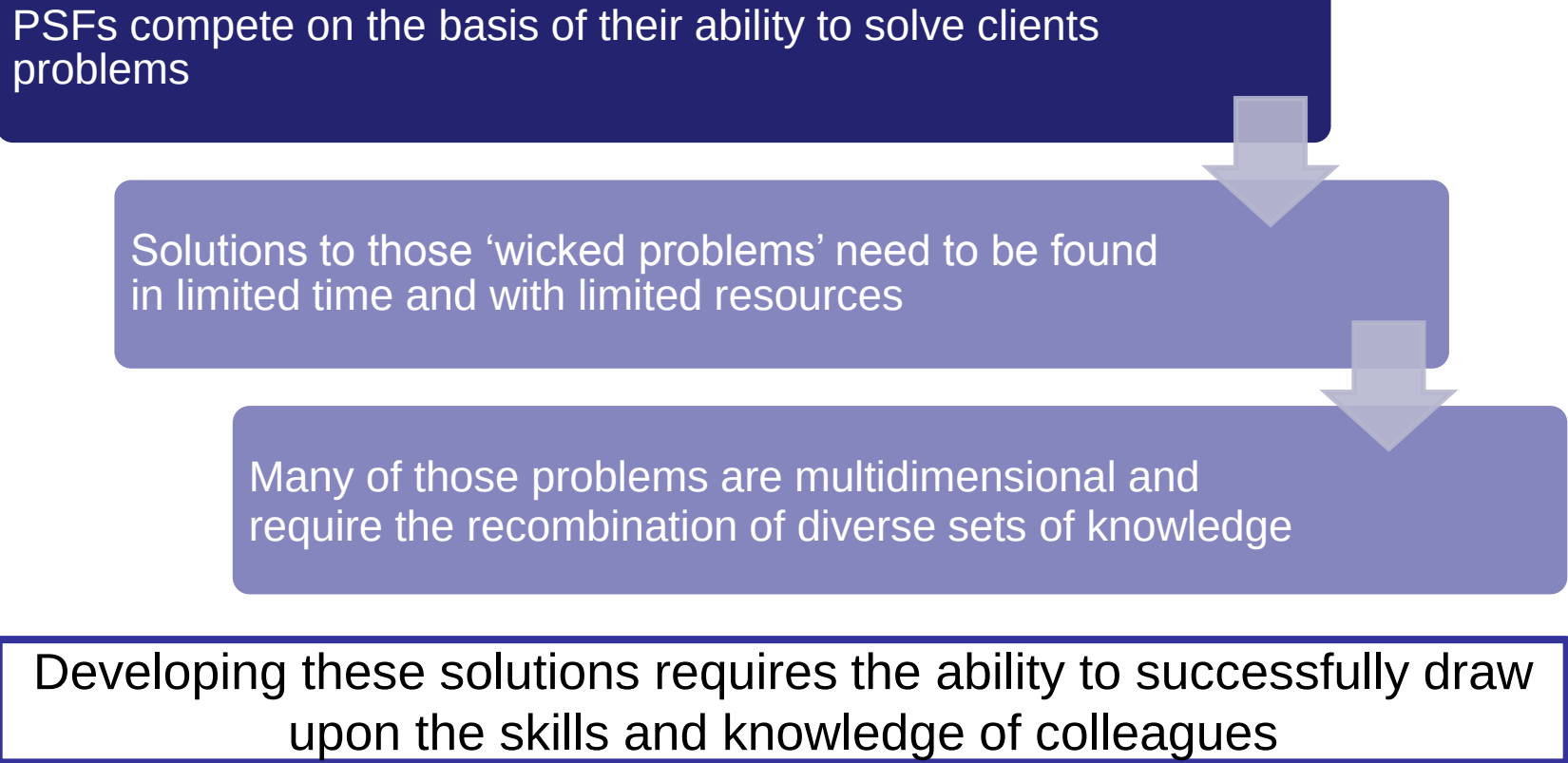
flying fox
relocation

Source: Criscuolo, Salter & Ter Wal (2013, WP)

Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS

PSFs compete on the basis of their ability to solve clients problems



```
graph TD; A[PSFs compete on the basis of their ability to solve clients problems] --> B[Solutions to those 'wicked problems' need to be found in limited time and with limited resources]; B --> C[Many of those problems are multidimensional and require the recombination of diverse sets of knowledge]; C --> D[Developing these solutions requires the ability to successfully draw upon the skills and knowledge of colleagues];
```

Solutions to those 'wicked problems' need to be found in limited time and with limited resources

Many of those problems are multidimensional and require the recombination of diverse sets of knowledge

Developing these solutions requires the ability to successfully draw upon the skills and knowledge of colleagues

Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS

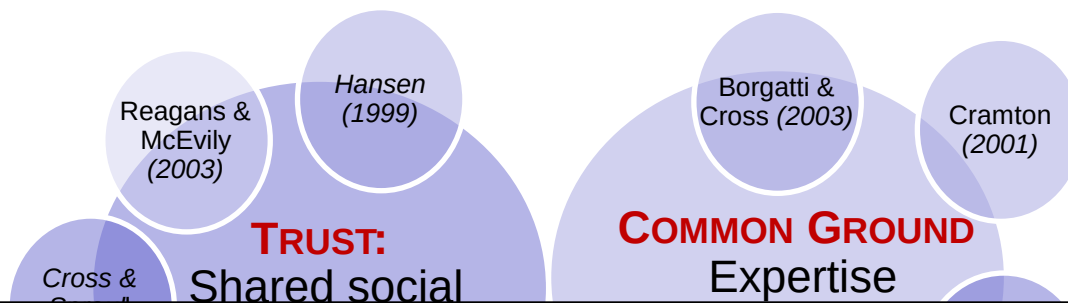
How can we explain patterns of knowledge sharing among consultant engineers?

...with diverse **skills** and **expertise**

...with diverse **prior work experience**

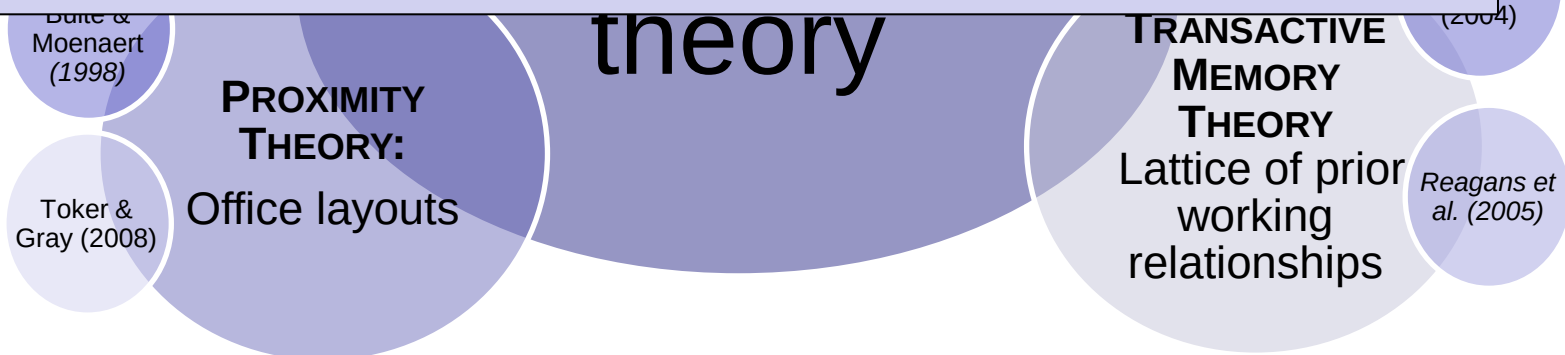
...spread over **various offices** in a single city

...with **shared friendship relations** among them?



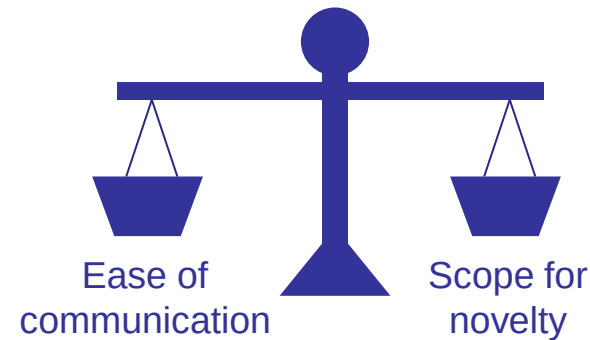
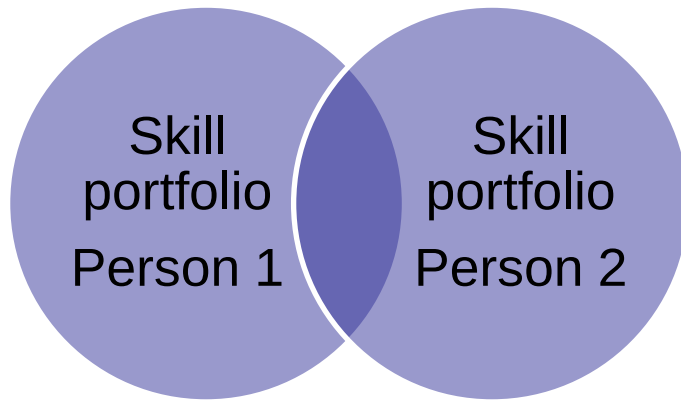
“Despite that various research traditions have advanced our understanding of organizational networks, there has been little effort to understand how various mechanisms interact”.

Nebus (2006)

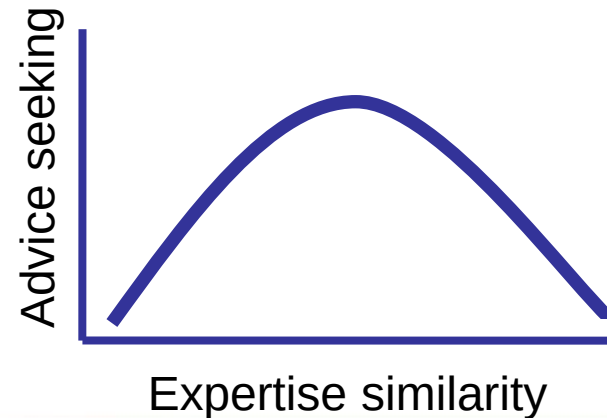


Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS

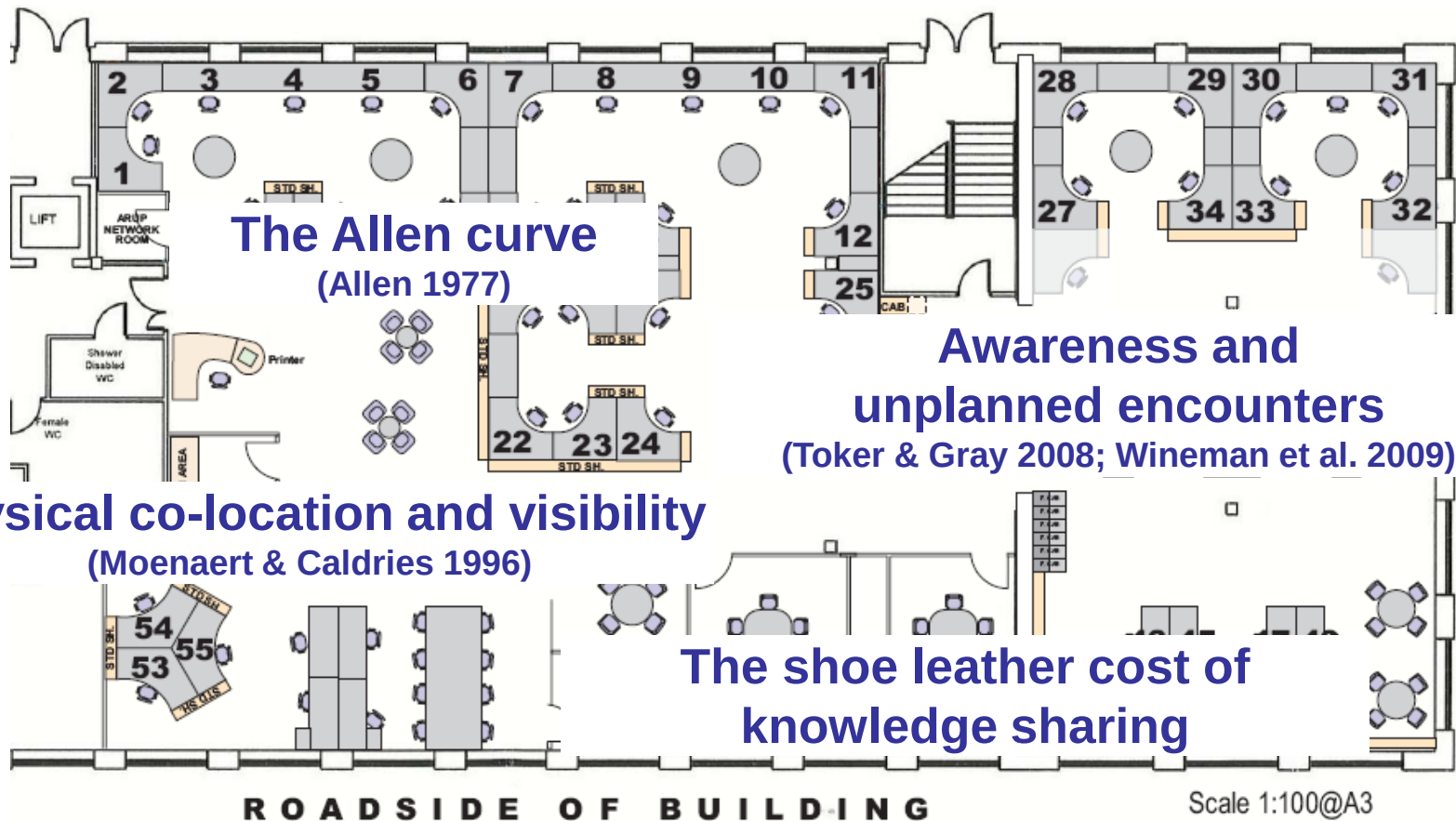


Balancing
ease of communication
and
scope for novelty



Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS



Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS



**Shared work
experience**

**Shared work
practices**

**Geo-
engineer**

**Consultancy
project**

**Acoustics
expert**

**Risk
manager**

**Developing
transactive memory**

Source: Criscuolo, Salter & Ter Wal (2013, WP)

Intra-organizational networks

EXOGENOUS DYADIC MECHANISMS

**Transitivity,
reputation and
commitment**
(Coleman 1990)



**Social cohesion
around a relation**
(Reagans and McEvily 2003)

Balance theory
(Heider 1946)

**Swift trust and trust
brokerage**
(Shapiro 1987)

Source: Criscuolo, Salter & Ter Wal (2013, WP)

Learning objectives

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Implications for your research

- **Context** is important for the study of tie formation. Tie formation mechanisms are widely applicable, but no one-size-fits-all.
- Mechanisms manifest themselves in a **complex interplay** of factors.
- Endogenous and exogenous mechanisms are both important: watch out for taking an **overly structuralist** perspective on networks.
- Paper must be context-specific, but advisable to pick one or more **theories/mechanisms to link into**.
- Any paper should **control for established mechanisms** that have been shown to have an impact on tie formation.

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